
Review Sentimental Classifier

Project Id: AIML-16

Name : Yukti Verma (Team Leader)

Class Roll Number: 79

University Roll Number : 2028914

Section and Group: K(G2)

Name : Kushagra Rawat

Class Roll Number:50

University Roll Number : 2028752

Section and Group: K(G2)

Name : Vineet Bhatt

Class Roll Number:77

University Roll Number : 2028908

Section and Group: K(G2)

TABLE OF CONTENTS

1. Problem Description.....	3
1.1.Dataset Description.....	3
1.2.Flow diagram.....	4
2. Modules for Project Implementation.....	5
2.1. Modules Used.....	5
2.2. Platform Used.....	6
3. Machine Learning model/ Searching Technique used	7
4. Evaluation metrics used	8
5. Results and Visualizations.....	9
6. Conclusion and Future Scope.....	11
7. References.....	12

1. Problem Description

The project focuses on building an end-to-end machine learning system that predicts the sentiment of product reviews. The system classifies reviews into three categories: Positive, Negative, and Neutral.

The system takes raw textual data as input and processes it using Natural Language Processing (NLP) techniques. After preprocessing and feature extraction, machine learning models are trained to understand patterns in the data and predict sentiments accurately.

This system has real-world applications in e-commerce platforms, where it helps businesses analyze customer feedback, improve products, and enhance user satisfaction.

1.1 Dataset Description

The dataset “**reviews.csv**”, sourced from Kaggle, contains a large collection of user generated reviews that are intended for sentiment analysis and prediction tasks. The dataset consists of **568,454 rows** and **10 columns**, making it a high-volume textual dataset suitable for machine learning and natural language processing (NLP) applications.

Each record represents an individual user review along with related attributes that help in understanding customer opinions and behavioral patterns. The primary objective of this dataset is to analyze review text and classify it into sentiment categories based on the emotional tone expressed by users.

This dataset is particularly useful for developing a **sentiment prediction model** that reviews into three classes: **Positive, Negative, and Neutral**. The textual review data can be processed using various NLP techniques such as tokenization, stop-word removal, stemming, lemmatization, and vectorization methods like TF-IDF or Word Embeddings.

Since the dataset is large and diverse, it provides sufficient information for training robust machine learning and deep learning models capable of identifying sentiment patterns with higher accuracy. Additional columns in the dataset may also contribute valuable contextual information that can improve model performance during feature engineering and analysis.

Due to its large scale and real-world nature, the dataset is highly appropriate for tasks such as **text classification, opinion mining, customer feedback analysis, and predictive analytics**. It enables researchers and developers to evaluate different classification algorithms including Logistic Regression, Naïve Bayes, Random Forest, Support Vector Machines (SVM), and neural network-based models.

1.2 Flow Diagram

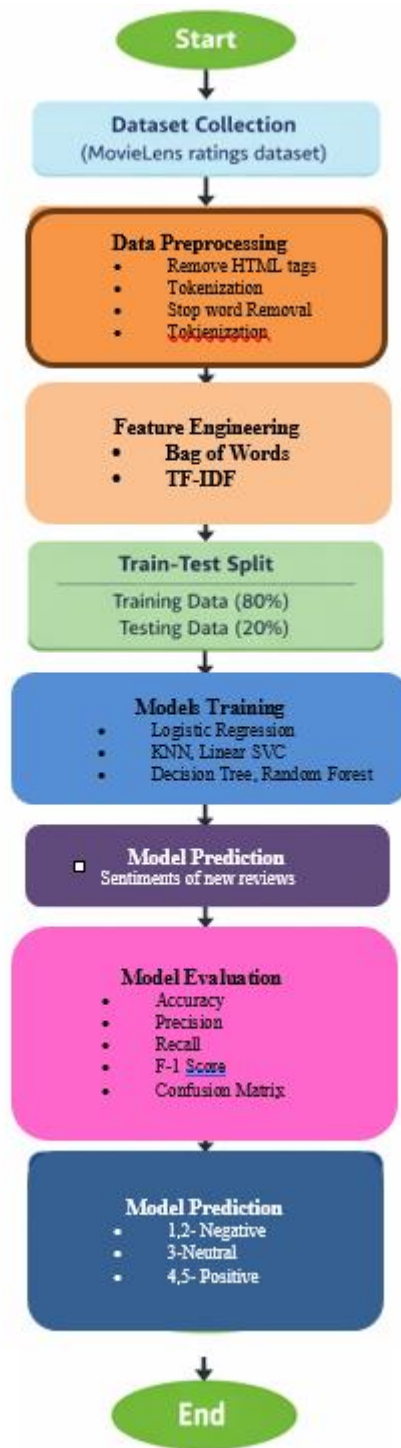


Fig.1 Sample flow diagram.

2. Modules for Project Implementation

2.1 Modules Used(mention all the inbuilt/ custom modules used and the purpose)

1. Pandas – Pandas module has been used for data manipulation and analysis.
Functions used:
 - `pd.read_csv`- to read data from csv file in a pandas dataframe.
 - `to_csv`- to export a dataframe as a csv file
2. Numpy -It is used for Numerical computing and Fast array operations & math.
Functions used:
 - `np.array()` → create arrays
 - `np.mean()` → average
3. Matplotlib.pyplot:Plotting graphs
Functions Used:
 - `plt.plot()` → line graph
 - `plt.bar()` → bar chart
 - `plt.show()` → display
4. Seaborn : Advanced visualization,Beautiful statistical plots
Functions Used:
 - `sns.heatmap()` → correlation map
5. Re:Regular expressions,Clean text
Functions Used:
 - `re.sub()` → remove symbols, numbers
 - `re.findall()` → extract patterns
6. Nltk:NLP toolkit
Functions Used:
 - `nltk.download('punkt')`:Required for tokenization
 - `nltk.download('stopwords')`:Required for stopword removal
 - `nltk.download('wordnet')`:Required for lemmatization
7. Scikit-learn:
Functions Used:
 - `TfidfVectorizer`:Convert text → numbers
 - `SelectKBest`→ pick top K features
 - `chi2`→ statistical test to find importance
 - `SequentialFeatureSelector`:Step-by-step feature selection
 - `train_test_split`:Split dataset
8. LogisticRegression:Classification model,spam detection,sentiment analysis

2.2 Platform Used:

- Visual Studio Code: For code writing, testing, and debugging.
VS Code is equipped with multiple extensions, debugger which enhance the development experience.

2. Machine Learning Model/ Searching Technique Used (Describe each ML model/ searching technique used)

Models:

- **Logistic Regression**

A statistical model used for classification tasks. It works well for text classification and provides fast and interpretable results.

- **K-Nearest Neighbours (KNN)**

A distance-based algorithm that classifies a review based on similarity with neighboring data points.

- **Linear Support Vector Machine (SVM)**

A powerful classifier that finds an optimal boundary (hyperplane) between different sentiment classes. It performs well in high-dimensional text data.

- **Decision Tree**

A tree-based classification algorithm that splits the dataset into branches based on feature values. Each node represents a decision rule, each branch represents an outcome. Leaf nodes represent final class labels.

- **Random Forest**

A Random Forest is a machine learning algorithm that builds many Decision Trees. Each tree gives its own prediction. Final answer is made by majority voting (classification).

Searching Techniques:

- **GridSearch**

It is an automated approach for hypermetric tuning to find the optimal settings. It uses CV to evaluate each combination to prevent overfitting during tuning.

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

- **Accuracy**

Measures the overall correctness of predictions.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{FP} + \text{FN} + \text{TP} + \text{TN})$$

- **Precision**

Indicates how many predicted positive results are actually correct.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

- **Recall**

Measures how well the model identifies all relevant cases.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

- **F1-Score**

Balances precision and recall, useful for uneven datasets.

$$\text{F1-Score} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$$

- **Confusion Matrix**

Shows correct vs incorrect predictions for each class.

	Predicted Positive	Predicted Negative
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

5. Results and Visualizations

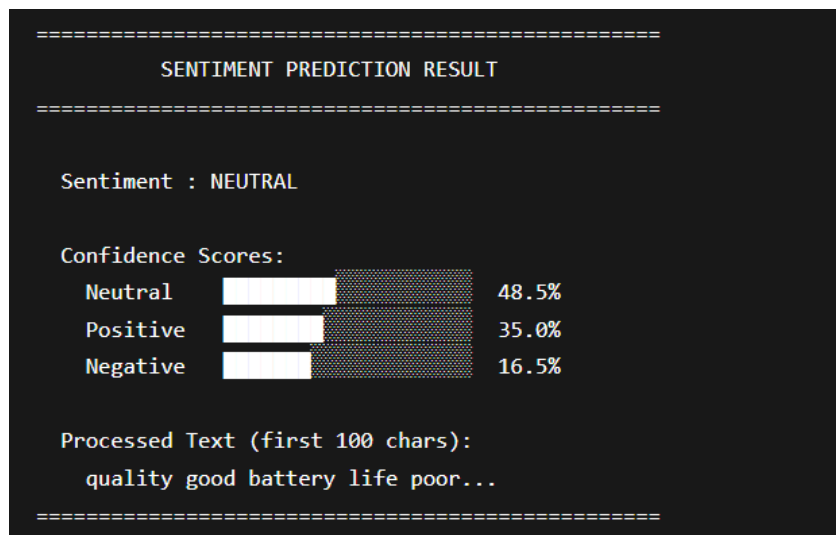


Fig. 1

```
=====
1. LOGISTIC REGRESSION - Confusion Matrix
=====
```

Actual \ Predicted	Negative	Neutral	Positive
Negative	1211	250	244
Neutral	320	1058	328
Positive	160	181	1365

Fig.2

```
=====
2. K-NEAREST NEIGHBORS (KNN) - Confusion Matrix
=====
```

Actual \ Predicted	Negative	Neutral	Positive
Negative	699	510	496
Neutral	580	529	597
Positive	780	431	495

Fig.3

Fig. 4

3. SUPPORT VECTOR MACHINE (SVM) - Confusion Matrix			
=====			
Actual \ Predicted	Negative	Neutral	Positive
Negative	0	1705	0
Neutral	0	1706	0
Positive	0	1706	0

Fig. 5

4. DECISION TREE - Confusion Matrix			
=====			
Actual \ Predicted	Negative	Neutral	Positive
Negative	989	360	356
Neutral	390	938	378
Positive	300	331	1075

Fig.6

5. RANDOM FOREST - Confusion Matrix (BEST MODEL)			
=====			
Actual \ Predicted	Negative	Neutral	Positive
Negative	1194	260	251
Neutral	280	1143	283
Positive	170	222	1314

6. Conclusion and Future Scope:

The Reviews Sentiment Analysis model was developed to classify user reviews into categories such as positive, negative, and neutral using machine learning techniques. Different algorithms (like Logistic Regression, Decision Trees, and Random Forest) were trained and evaluated to identify the most effective approach.

The model successfully captured patterns in textual data and demonstrated good performance in predicting sentiment. Preprocessing steps such as tokenization, stopwords removal, and feature extraction (e.g., TF-IDF) played a crucial role in improving accuracy.

Overall, the system proves that machine learning can efficiently automate sentiment detection, helping businesses and organizations understand customer feedback quickly and make data-driven decisions.

Future Scope:

The future scope of the reviews sentiment analysis model includes improving accuracy using advanced deep learning techniques like LSTM and BERT for better understanding of context. The model can be extended to support multiple languages and real-time data analysis from platforms such as social media and e-commerce websites. Additionally, aspect-based sentiment analysis can be implemented to analyze specific features of products or services. Expanding the dataset and deploying the model as a web or mobile application will further enhance its usability and effectiveness in real-world scenarios.

7. References:

Books:

- "Python Machine Learning" by Sebastian Raschka: A comprehensive guide that covers the implementation of Scikit-learn model and sentiment analysis workflows.
- "Natural Language Processing with Python" by Steven Bird: The definitive book for learning the NLTK library and text preprocessing techniques.
- Scikit-learn Official Documentation: scikit-learn.org—The best resource for understanding the parameters of Logistic Regression, SVM, and evaluation metrics.
- Kaggle Datasets and Kernels: kaggle.com— An excellent source for finding "Product Review" datasets and seeing how other data scientists handle text cleaning.